

UNIVERSIDAD DE SEVILLA
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MÁSTER UNIVERSITARIO EN MICROELECTRÓNICA:
DISEÑO Y APLICACIONES DE SISTEMAS MICRO/NANOMÉTRICOS

**Design of a Key-Generation Cryptographic
Engine for Integrated-Circuit
Implementation: a Transistor-Level
Characterized Entropy Source with FPGA
Verification against an ESP32**

Master's Thesis

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Abstract

Cryptographic keys are only as good as the entropy behind them, yet most embedded key generators are evaluated as black boxes: statistical test suites are run on their output and a pass is taken as evidence of randomness. The author's previous work on the hardware random number generator of the ESP32 showed why that is not enough — a source with low min-entropy can pass NIST SP 800-22 — and NIST itself has since rejected the suite for assessing cryptographic generators. This thesis takes the opposite route: a white-box key-generation engine whose entropy is justified from the physics of the device. A ring-oscillator entropy source is designed at transistor level and its jitter is extracted from simulation and turned, through a validated stochastic model, into a lower bound on min-entropy per bit and into the design parameters of the sampler. Around it, a fully portable digital core — SP 800-90B health tests, an AES-based conditioning and CTR_DRBG with derivation function, AES-CMAC authentication and an I2C register interface — is written once and verified three ways: against the official test vectors, in simulation on a Basys 3 FPGA acting as the verification vehicle, and against an ESP32 that recomputes every result with an independent implementation (mbedtls). The same VHDL is finally synthesized on an open 130 nm standard-cell library to give the area in gate equivalents and the path to a dedicated integrated circuit. All code, models, vectors and data are released.

Keywords true random number generator, ring oscillator, jitter, min-entropy, stochastic model, AIS 20/31, SP 800-90B, CTR_DRBG, AES-CMAC, secure element, FPGA, ASIC, ESP32.

Resumen

Las claves criptográficas valen lo que vale la entropía que hay detrás, y sin embargo la mayoría de los generadores embebidos se evalúan como cajas negras: se pasan baterías de tests estadísticos sobre su salida y un aprobado se toma como prueba de aleatoriedad. El trabajo previo del autor sobre el generador hardware del ESP32 mostró por qué no basta — una fuente con poca min-entropía puede pasar NIST SP 800-22 — y el propio NIST ha rechazado después esa batería para evaluar generadores criptográficos. Este TFM toma el camino contrario: un motor de generación de claves de caja blanca cuya entropía se justifica desde la física del dispositivo. Se diseña a nivel de transistor una fuente de entropía por oscilador de anillo, se extrae su jitter por simulación y se convierte, mediante un modelo estocástico validado, en una cota inferior de min-entropía por bit y en los parámetros de diseño del muestreador. A su alrededor, un núcleo digital totalmente portable — tests de salud SP 800-90B, acondicionado y CTR_DRBG con función de derivación basados en AES, autenticación AES-CMAC e interfaz de registros I2C — se escribe una sola vez y se verifica de tres maneras: contra los vectores oficiales, en simulación sobre una FPGA Basys 3 que hace de vehículo de verificación, y frente a un ESP32 que recalcula cada resultado con una implementación independiente (mbedtls). El mismo VHDL se sintetiza por último sobre una biblioteca abierta de celdas estándar de 130 nm para dar el área en puertas equivalentes y el camino hacia un circuito integrado dedicado. Todo el código, modelos, vectores y datos se publican.

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List of Acronyms

AES	Advanced Encryption Standard
AIS	Anwendungshinweise und Interpretationen zum Schema (BSI)
APT	Adaptive Proportion Test
ASIC	Application-Specific Integrated Circuit
BSI	Bundesamt für Sicherheit in der Informationstechnik
CAVP	Cryptographic Algorithm Validation Program
CDC	Clock Domain Crossing
CMAC	Cipher-based Message Authentication Code
CTR_DRBG	Counter-mode Deterministic Random Bit Generator
DRBG	Deterministic Random Bit Generator
ERO	Elementary Ring Oscillator
FPGA	Field-Programmable Gate Array
GE	Gate Equivalent
I2C	Inter-Integrated Circuit
KAT	Known-Answer Test
LUT	Look-Up Table
MEMS	Micro-Electro-Mechanical System
MPW	Multi-Project Wafer
NIST	National Institute of Standards and Technology
PDK	Process Design Kit

PLL	Phase-Locked Loop
PUF	Physically Unclonable Function
RCT	Repetition Count Test
RO	Ring Oscillator
RTL	Register-Transfer Level
SPI	Serial Peripheral Interface
TRNG	True Random Number Generator
XADC	Xilinx Analog-to-Digital Converter

Chapter 1

Introduction

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7.3 Comparison with commercial secure elements

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Chapter 8

Conclusions and Future Work

8.1 Conclusions

8.2 Contributions

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Appendix A

Register Map

Appendix B

Official Test Vectors Used

Appendix C

Reproduction Guide